JONATHAN NADEAU

DATA ANALYST

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Dear Hiring Manager,

I was a bright, curious, and active kid. I spent my seventh grade afternoons binging Vsauce, MinutePhysics, and SciShow before soccer practice, wondering about questions like what shape our field of vision is. I excelled in class and later in soccer, training my way up to a premier level club team. Before college, I was good in STEM.

I got into UCSB for computer engineering. My favorite class was the ECE 10A lab, where I could talk in more depth with the teacher about circuit components and designs. COVID made it harder for me to perform in my engineering classes, and I ended up changing majors to mathematics. I connected the dots and realized I had to have a career in mind, so I chose teaching and minored in secondary math and science education, studying pure math and pedagogy while volunteer tutoring at a local high school. I graduated with four Dean's Honors in my final quarters.

After graduating I worked at a private school, tutoring middle and high school students and conducting classroom management, while also tutoring after hours for Blue Train College Prep and Wyzant, mainly math but also SAT prep. In my second year there I began the credential program at CSU Long Beach, took the classes seriously, and earned a 4.0 GPA across all coursework. I spent the summer before student teaching creating around a hundred lesson plans, anticipating Algebra 1, Geometry, and Algebra 2. Student teaching went better than I thought it would. I expected the kids would not listen to me at all. They did, for the most part. One called me the GOAT, and when the surveys came back from my Algebra 2 class, most if not all of them said my teaching was of good quality.

Then one of my mentor teachers asked me where I wanted to live. I said I would love to stay in the South Bay, and she told me that would be tough on a teacher's salary. That really made me think, and I started researching careers with more upward income mobility. Considering my mathematics bachelor's, I landed on data analytics, made the decision, and put all of my energy toward that path. I completed the Google Data Analytics Professional Certificate in two months with a 100% average across all nine courses, built my portfolio, and started applying for jobs.

I then reached out to my past employer, Escape Communications, to see whether they had analytics work I could take on, and they graciously offered me some. The first project was figuring out what to do with their old Bugzilla failure database. We scoped it together, seeing what the lab technician and the board designer wanted out of the tool while I researched what was possible. I pulled the bug data through a REST API from within the same network as the server it sits on: 4,038 failure records spanning 2007 through 2026, with 13,252 engineer comments. I parsed the summaries and comment text for reference designators, going back and forth refining the parsing algorithms to minimize the false positive rate, validating eleven component prefixes by hand and dropping one after sampling put its rate near 85%. We wanted the reference designators because those are specific locations on a board, so knowing that a set of failures share components is exactly what the lab tech and the board guy want to know. I built them a Streamlit dashboard where they can search bugs across every available data field, faster than Bugzilla's own search, and added a TF-IDF similarity implementation that surfaces bugs similar to a selected one based on the key terms in their text entries. We mounted the app on a company server, where it now runs connected to the Bugzilla source and refreshes for all new bugs at the press of a button.

My second project was a BOM checker. It takes in the bill of materials exactly as exported from the PADS Logic board design system, decodes the part numbers to determine what the other values should be, compares that against what was entered in the other fields, and flags the inconsistent ones, catching upstream errors made inside the schematic layout tool. Validated across eight production boards, nine of ten flagged mismatches turned out to be genuine errors against the manufacturers' own datasheets. Next was the BOM sourcing tool, which takes in BOMs and pulls availability and pricing from DigiKey and Mouser through their search APIs, so the VP of Engineering can quickly find out what parts he needs to order and from where, or whether they need to redesign around parts they cannot obtain in time for a production contract.

The next tool was the ILA waveform file checker, for the FPGA engineer. He told me he usually verifies his loopback modulation waveforms by eyeballing them to see whether they look the same, and I told him I could make a tool that verifies it through Python processing instead. It takes in his capture, tells him the characteristics of the waveforms in the file, and reports any input or output discrepancies down to the exact word and sample index. After that came an inventory audit log tool, which lets the VP keep track of the parts they audit for, their counts, and the dates those counts were made, addressing a difficulty they face every audit from the messy and undocumented data that falls through the cracks of their PEDYN inventory protocols. The latest is a PEDYN table data reconciliation and report generating tool, which outputs the Excel files they need during audit season: COGS, on time delivery, production yields, resource procurement, and more. Reverse-engineering six of those undocumented reports from twenty archived workbooks and sixteen ERP table schemas let me reproduce four consecutive years of historical output exactly and surface four material errors carried in the existing record.

In this work I have been able to learn their operations and systems, work with them to identify room for improvement, and then build the tools that make those improvements. I feel like I have developed a keen eye for optimization potential. All of my projects involve data, and I am familiar with different forms of data and file types and what can be deduced from analyzing them. I am eager to land a role where I can contribute even more company value, and eager to put in the work to serve and earn my living.

Sincerely,

Jonathan Nadeau